



## Protocol name: Post-staining easyIRF sections for electron microscopy

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Abbreviations: IRF - In Resin Fluorescence; UA - Uranyl Acetate; ITO - Indium Tin Oxide

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### Health & Safety requirements

Ensure you are familiar with all relevant local Health and Safety documents and procedures (Material Safety Data Sheets, Procedural Risk Assessments, Standard Operating Procedures etc.) before starting.

**Safety critical:** All procedures designated safety critical must be performed in a fume hood (where appropriate) and the following PPE must be worn - lab coat, gloves, and eye protection.

**Procedural Risk Assessments:** Handling of uranyl acetate, lead citrate, potassium permanganate, sodium hydroxide

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### Materials

- ☐ 15 ml glass scintillation vials with caps
- ☐ 2 ml cryovial
- ☐ Hot plate set to 60°C
- ☐ Reverse fine tip tweezers
- ☐ Glass slide
- ☐ Plastic box with lid
- ☐ Parafilm
- ☐ Glass beakers
- ☐ Aluminium foil
- ☐ Blotting paper
- ☐ Aluminium pin SEM stubs (check compatibility with microscope) Agar AGG301
- ☐ Sticky carbon pad Agar AGG3347N
- ☐ Silver paint Agar AGG3691

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### Chemicals

- ☐ ddH<sub>2</sub>O; double distilled water

- ☐ Potassium permanganate
- ☐ Sodium hydroxide pellets

## Reagents

- ☐ Staining solution 1 - 2% uranyl acetate + 0.5% potassium permanganate
- ☐ Staining solution 2 - Reynold's lead citrate

### Staining solution 1

*Stock solution: 4% uranyl acetate* **Safety critical**

Use Reagent Protocol RP01: Preparation of aqueous uranyl acetate, adjusting the concentration to make 4% uranyl acetate.

|                |                 |                                            |                                                                                    |
|----------------|-----------------|--------------------------------------------|------------------------------------------------------------------------------------|
| Uranyl Acetate | State:<br>Solid | Hazard Codes:<br>H300, H330, H373,<br>H411 |  |
|----------------|-----------------|--------------------------------------------|------------------------------------------------------------------------------------|

*Stock solution: 1% potassium permanganate*

Weigh 0.1 g of  $\text{KMnO}_4$  powder into a 15 mL glass vial

Add 10 ml ddH<sub>2</sub>O and mix well

Close cap and seal with parafilm to prevent oxidation

Store at 4°C

|                        |                 |                                                                  |                                                                                      |
|------------------------|-----------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Potassium Permanganate | State:<br>Solid | Hazard codes:<br>H272, H302, H314,<br>H318, H361d, H400,<br>H410 |  |
|------------------------|-----------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------|

*Working solution: 2% uranyl acetate + 0.5% potassium permanganate* **Safety critical**

Add 1 ml of 4% uranyl acetate to a 2 ml cryovial

Freeze at -20°C

Add 1 mL of 1%  $\text{KMnO}_4$  to the frozen uranyl acetate and thaw at RT

Once thawed, vortex the mixture

### Staining solution 2

Use Reagent Protocol RP02: Preparation of Reynold's Lead Citrate **Safety critical**

|              |                  |                                                    |                                                                                      |
|--------------|------------------|----------------------------------------------------|--------------------------------------------------------------------------------------|
| Lead Citrate | State:<br>Liquid | Hazard Codes:<br>H302, H332, H360Df,<br>H373, H410 |  |
|--------------|------------------|----------------------------------------------------|--------------------------------------------------------------------------------------|

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## Procedure

1. Prepare Staining Solution 1 and Staining Solution 2. **Safety critical**
2. Set and heat the hotplate to 60°C.
3. Label the tweezers and a clean glass slide with the sample name.
4. Place the ITO coverslip with the sections facing upwards on the glass slide and then place the slide onto the hotplate for 10 mins.

Note: This baking step prevents the sections from detaching from the ITO coverslip during the staining and washing steps.

5. Line the lid of the plastic box with parafilm and place one drop of Staining Solution 1 and one drop of ddH<sub>2</sub>O onto the parafilm for each coverslip to be stained (Fig.1A). **Safety critical**

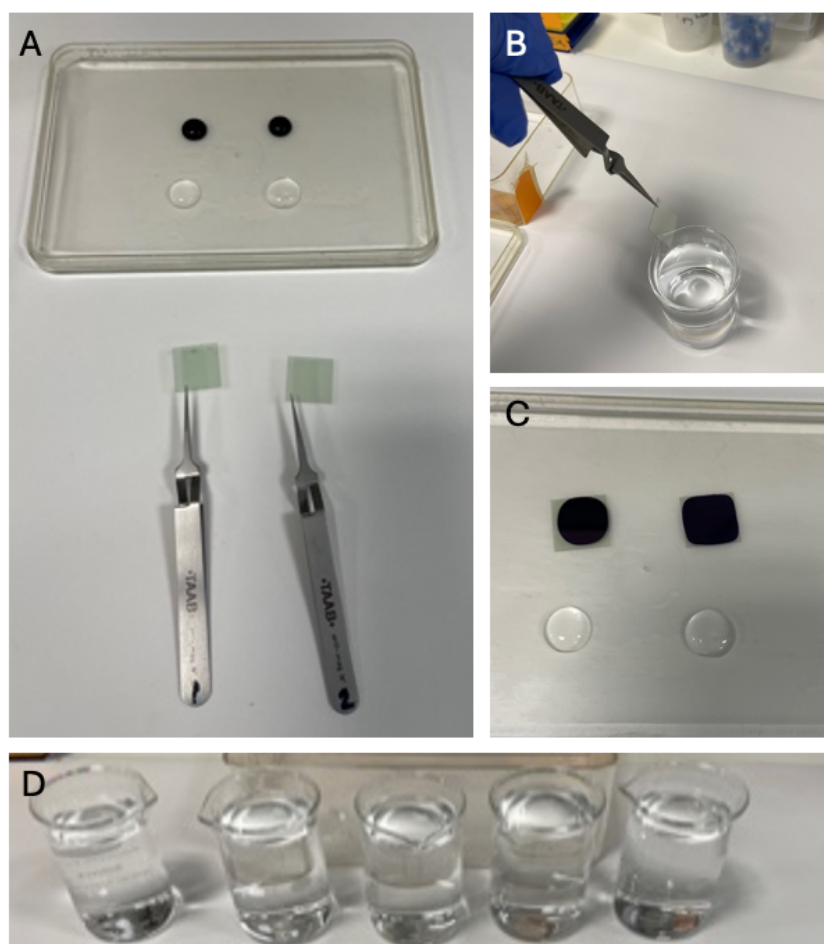


Figure 1. Layout for staining and washing of ITO coverslips.

6. Wet the ITO coverslip by dipping it in a glass beaker of ddH<sub>2</sub>O (Fig.1B).

7. Place the ITO coverslip onto the drop of stain with the sections facing down (the ITO should float on the drop). Repeat step 6 and 7 for the remaining ITO coverslips (Fig.1C).
8. Leave ITO coverslip(s) on the stain for 10 mins.
9. Fill 5 glass beakers with ddH<sub>2</sub>O (Fig.1D).
10. After 10 mins, carefully lift the ITO coverslip and place onto the drop of water. Repeat for remaining ITO coverslips.
11. Carefully lift the ITO coverslip from the drop of water and rinse by dipping ten times vertically in the first glass beaker of ddH<sub>2</sub>O, then repeat for the remaining four beakers to complete 50 x fast washes. Repeat this step for the remaining ITO coverslips.
12. Blot the coverslips dry at the edge, being careful not to touch the sections or deposit lint onto the glass.

Note: Dispose of mixed uranyl acetate and potassium permanganate waste according to local rules.

13. Exchange the parafilm lining the lid and place one drop of Staining Solution 2 and one drop of ddH<sub>2</sub>O onto the parafilm for each coverslip to be stained (Fig.1A). Add sodium hydroxide pellets close to the stain to absorb carbon dioxide and prevent precipitation of the lead. **Safety critical**

|                  |               |                      |                                                                                       |
|------------------|---------------|----------------------|---------------------------------------------------------------------------------------|
| Sodium Hydroxide | <b>State:</b> | <b>Hazard Codes:</b> |  |
|                  | Solid         | H290, H314           |                                                                                       |

Note: Work quickly and minimize exposure of the stain drops to carbon dioxide. Do not breathe onto the drops.

14. Repeat Steps 7-12.
15. Leave the coverslips to dry in the reverse forceps overnight, protected from dust and in the dark.

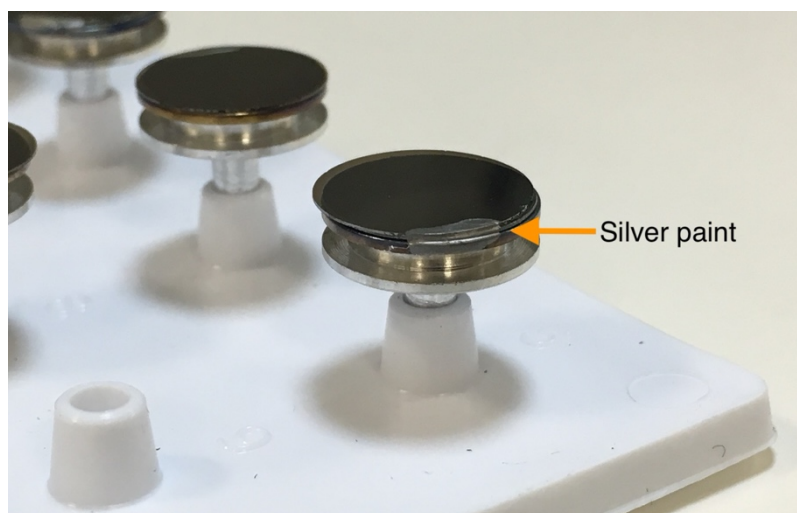


Figure 2. Mounting coverslips on SEM stubs.

16. Put the silver paint bottle on a roller mixer to resuspend for 5 mins.

|                           |                         |                                                   |                                                                                                                                                                          |
|---------------------------|-------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quick Drying Silver Paint | <b>State:</b><br>Liquid | <b>Hazard Codes:</b><br>H225, H302, H319,<br>H335 |   |
|---------------------------|-------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

17. Place an adhesive carbon tab on an SEM stub and place the ITO coverslip on it with the sections facing up.
18. Paint part of the edge of the coverslip and stub with silver paint so there is a conductive path from the glass to the metal of the stub (Fig.2).